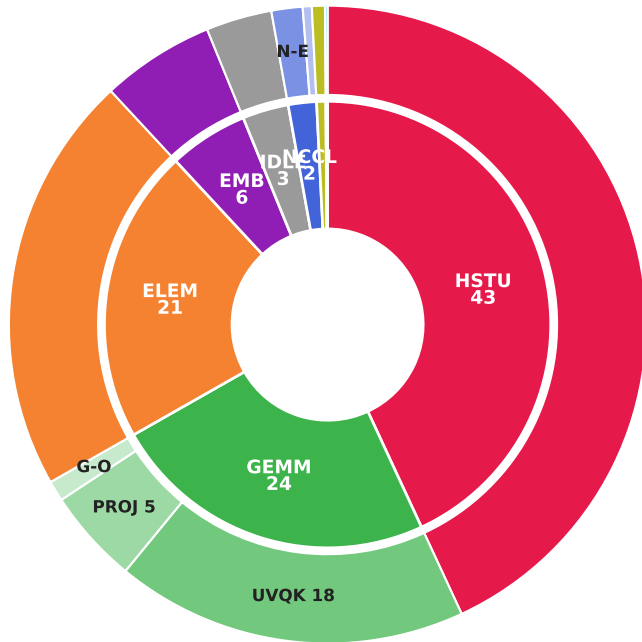
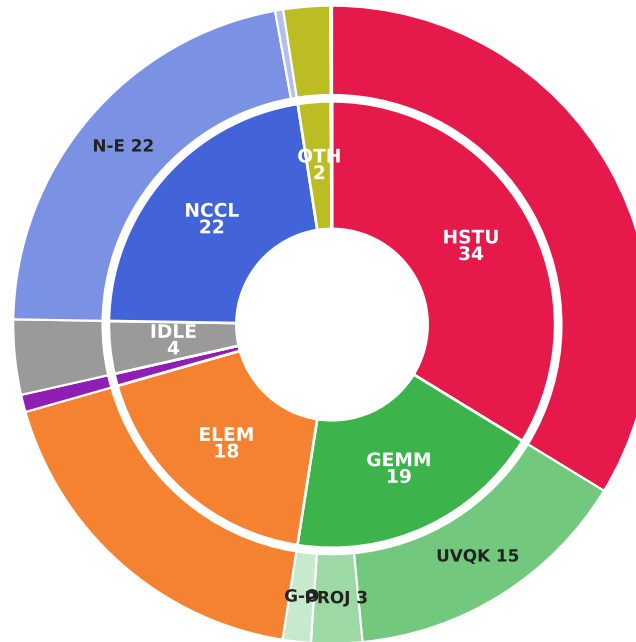


HSTU GPU-time sunburst — NON-JAGGED (matched config), exposed accounting, fastest step (upstream \$2.2 method)  
 inner ring = grouping · outer ring = leaves · % of the step

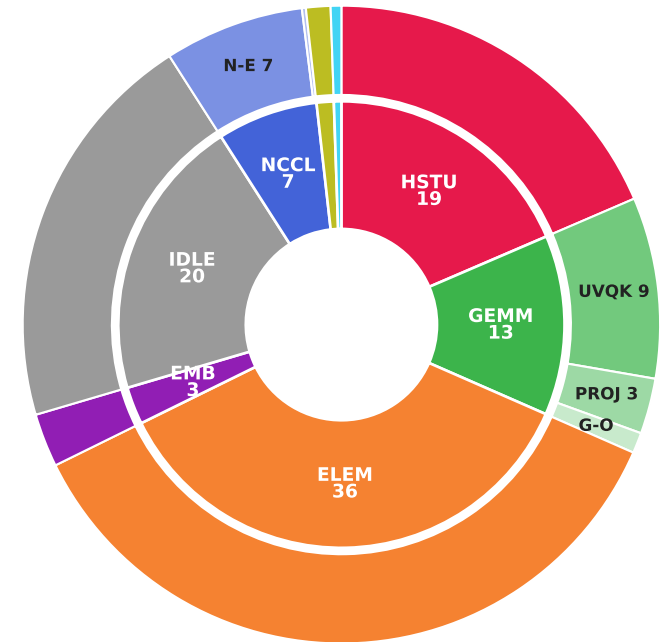
Upstream H100 · step 162 · D256 (nj)



Our H100 · step 159 · D256 (nj)



Our GB300 · step 153 · D128 (nj)



■ HSTU = hstu fwd/bwd   
 ■ PROJ = gemm / projection   
 ■ GEMM = gemm (ours: not sub-split)   
 ■ EMB = embedding op   
 ■ IDLE = GPU idle   
 ■ OVL = overlapped  
■ UVQK = gemm / uvqk   
■ G-O = gemm / others   
■ ELEM = elementwise   
■ NCCL = nccl (exposed)   
■ OTH = others